

James Gaboriault-Whitcomb

☎ xxx-xxx-xxxx | ✉ xxxxxxxxxx@gmail.com | 🌐 github.com/jjdoesit

Education

Youngstown State University

Bachelor of Science in Computer Science | 3.95 GPA

Expected May 2027

Youngstown, OH

Relevant Coursework

Programming and Problem Solving, Data Structures and Objects, Advanced Object-Oriented Programming, Development of Databases, Software Engineering

Work Experience

Software Developer Intern

Union Home Mortgage

May 2025 – Aug. 2025

Strongsville, OH

- Rewrote and modernized an internal legacy application ("Disaster Services") used to monitor active loans in federally declared disaster areas, significantly improving scalability and performance
- Developed a full-stack solution using C#, .NET Framework, Blazor, and SQL Server within a 5-tier layered architecture and RESTful API design
- Collaborated in a full Agile development cycle, including daily standups, sprint planning, estimation, and retrospectives

Machine Learning Research Assistant

Lawrence Berkeley National Lab

May 2024 – July 2024

Berkeley, CA

- Scaled and optimized a high-performance track particle reconstruction graph neural network (GNN) pipeline using data from CERN, doubling inference and training speeds
- Developed a visualization pipeline to stream output data into properly formatted plots, tables, and directories utilizing SciPy, Matplotlib, and Hydra, enhancing data analysis efficiency
- Delivered ongoing research and results to the Department of Energy, and presented findings at Indico Conference and IEEE International Conference on Big Data

Projects

2D Physics Engine | Java, LibGDX

- Designed a 2D physics engine supporting both rigid and soft bodies with customizable properties
- Achieved precise collision detection using the Separating Axis Theorem (SAT) and realistic impulse-based physics resolution
- Optimized performance using advanced techniques including spatial partitioning (Quadrees) for scalable collision handling

Spritesheet Lab | Next.js, React, MongoDB

- Developed a user-centric website for royalty-free sprite sheets, attracting 10 active users
- Implemented session-driven account registration and login processes, and email-based two-factor authentication
- Designed an advanced gallery page with MongoDB for enhanced search and filtering functionality
- Successfully deployed and hosted the website at spritesheetlab.com

Technical Skills

Languages: C#, Python, Java, C++, JavaScript, HTML/CSS, SQL

Frameworks/Technologies: Next.js, React, .NET, SQL Server Management Studio, MongoDB, Git